

ELIE GREG BANGA-BOTHY

Software Engineering Student | Backend & Fintech Engineering | Freelance
Web Developer

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GitHub: github.com/greglie498

PROFESSIONAL PROFILE

Software Engineering student (expected 2027) and freelance web developer with a backend-leaning full-stack profile and a focus on African fintech and mobile-money systems. Built **PanWallet**, a multi-provider mobile-money wallet integrating M-Pesa Daraja and MTN MoMo with idempotent transaction handling and hardened token-based authentication. Currently building a client website with Next.js and Supabase and developing **SkillLink**, a marketplace connecting students and early-career talent with SMEs, as a capstone project. Works across Node.js/TypeScript, React/React Native, PostgreSQL and Prisma. Fluent in English and French.

EDUCATION

United States International University–Africa

Bachelor of Science in Software Engineering

Nairobi, Kenya | January 2023 – Present | Expected graduation: 2027

Relevant academic areas include:

- Software Engineering
- Software Construction and Development
- Software Project Planning and Management
- Human-Computer Interaction
- Computer and Network Security
- Software Engineering Project
- Object-Oriented Programming
- Database Systems
- Systems Analysis and Design
- Mobile Application Development

Expected Graduation: 2027

TECHNICAL SKILLS

Programming Languages

- Java
- JavaScript
- TypeScript
- SQL
- HTML5
- CSS

Backend Development

- Node.js
- Express.js
- Spring Boot
- REST API development
- API integration
- Server-side application development
- Business logic implementation
- Controller-Service-Repository architecture
- DAO and Service Layer architecture

Frontend Development

- React
- React Native
- Expo
- HTML5
- CSS
- Tailwind CSS
- Responsive UI development
- Component-based development

Databases & Data

- PostgreSQL
- MySQL
- MongoDB
- SQLite
- Prisma ORM
- Database schema design
- CRUD operations
- Data validation
- Relational database design

Authentication & Security

- JWT authentication

- Firebase Authentication
- Phone OTP authentication
- Refresh token rotation
- Input validation
- Protected routes
- Role-based access control
- Prepared statements
- Secure API design

APIs & Integrations

- REST APIs
- M-Pesa Daraja API
- MTN Mobile Money API
- Exchange-rate API integration
- Third-party API integration
- Idempotency keys

Software Architecture & Engineering

- Clean Architecture
- Controller → Service → Repository architecture
- DAO / Service Layer architecture
- Object-Oriented Programming
- Requirements analysis
- Functional requirements
- Non-functional requirements
- Workflow design
- Technical documentation
- System design
- Database modeling

Development Tools

- Git
- GitHub
- Maven
- Postman
- VS Code
- Android Studio
- Expo
- Jest
- Supertest
- JUnit
- Docker

Engineering Practices

- Debugging
 - Problem solving
 - API testing
 - Unit testing
 - Integration testing
 - Code organization
 - Version control
 - Agile development
 - Collaborative development
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FREELANCE EXPERIENCE

Freelance Web Developer — Nini Assists

August 2026 – Present (in progress)

US-based boutique concierge business (event coordination, tutoring, translation) | Source code private (client work)

Technology: Next.js (App Router), TypeScript, Tailwind CSS v4, Supabase, Cloudflare, Vercel

- Scoping and building the client's business website, taking the engagement from requirements through wireframes to visual design.
 - Structured delivery in phases with a client sign-off gate before development to protect scope and payment.
 - Architecting on Next.js App Router with Supabase and Vercel/Cloudflare deployment; client-managed admin interface planned as a later phase.
 - [Add a concrete outcome once shipped: launch date, pages delivered, performance or lead metrics.]
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PROJECT EXPERIENCE

PanWallet — Pan-African Mobile Money Wallet

Backend / Full-Stack Project | 2026 – Present

Technology: Node.js, TypeScript, Express.js, PostgreSQL, Prisma, React Native, Expo, Firebase Authentication, M-Pesa Daraja API, MTN MoMo API, ExchangeRate API

PanWallet is a multi-provider digital wallet designed to address fragmentation within Africa's mobile-money ecosystem by providing a unified platform for wallet management and cross-network transactions.

Contributions

- Built a clean-architecture backend and mobile app module by module (10 backend and 10 mobile modules), using repository interfaces and dependency inversion to keep business logic independent of persistence.
- Integrated M-Pesa Daraja and MTN MoMo into a commons sandbox wallet service; replaced Orange Money after finding no unified pan-African API, and designed the provider layer to be extensible.
- Prevented duplicate transactions on unreliable mobile networks by enforcing idempotency on wallet operations.
- Secured sessions with JWT and hashed refresh tokens using family-based reuse detection, so a replayed token invalidates its whole token family.
- Standardized failures with machine-readable API error codes and validated environment configuration with Zod to fail fast on misconfiguration.
- Moved the project toward production: provisioned Neon staging and production Postgres branches, added staging-aware Prisma scripts, and logged and triaged defects in a tracked bug register.
- Wrote Jest/Supertest test suites for wallet and transaction services [N unit tests — run the suite and confirm the count before citing].
- Designed multi-currency conversion workflows using an exchange-rate service.

Repository: <https://github.com/greglie498/Pan-Wallet>

SkillLink – Student-SMEs Work Marketplace

Website Platform | 2026 (capstone, in progress)

Technology: React, TypeScript, Vite, Tailwind CSS, Node.js, Express, PostgreSQL, Prisma, JWT

- Designing a marketplace connecting university students and early-career professionals with SMEs for paid short-term projects, internships and apprenticeships.
- Authored the requirements and development plan: functional and non-functional requirements, user stories, and a deliberately scoped MVP (skill-based profiles, project postings, matching, applications, reviews), excluding escrow, disputes and native apps from the first release.
- Produced low-fidelity Figma wireframes and is converting them to a high-fidelity UI, with out-of-scope screens labeled as future phases.
- Planning a validated-workflow-first build, with frontend flows before the database and backend, and M-Pesa payments in a later phase.

Repository: <https://github.com/greglie498/SkillLink>

Lost & Found Application

Full-Stack Web Application | 2026

Technology: Node.js, Express.js, MongoDB, React, JWT, REST APIs

- Developed a full-stack Lost & Found platform for reporting, managing, searching, and claiming lost or found items.
- Implemented JWT-based authentication and protected application routes.
- Developed CRUD functionality for item records and reporting workflows.
- Built reusable React components for the application's user interface.
- Integrated RESTful APIs between the React frontend and Node.js/Express backend.
- Implemented MongoDB-based data persistence.
- Applied role-based access controls to protect user and administrative functionality.
- Designed the application to replace manual lost-and-found processes with a searchable digital workflow.

Repository: <https://github.com/greglie498/LostAndFound>

Employee Management System

Desktop Application | 2025

Technology: Java, Java Swing, MySQL, Maven, DAO, Service Layer, JCalendar, RS2XML

- Developed a Java Swing-based employee management application for managing organizational employee records.
- Implemented administrator authentication and protected management functionality.
- Developed CRUD operations for employee records, including adding, updating, deleting, and searching records.
- Integrated MySQL for persistent data storage.
- Applied DAO and Service Layer architecture to separate database operations from business logic.
- Used prepared statements for secure database interactions.
- Implemented searchable data views to improve employee-record retrieval.
- Used Maven to manage the project and external dependencies.
- Integrated JCalendar and RS2XML libraries to extend application functionality.

Repository: <https://github.com/greglie498/Employee-Management-System>

Student Management System

Android Application | 2025

Technology: Java, Android Studio, SQLite, XML

- Developed a mobile Student Management System for managing student records and academic information.
- Implemented student registration and record management functionality.

- Developed fee tracking and balance-calculation features.
- Implemented SQLite-based local data persistence.
- Designed multiple Android activities and XML layouts for application workflows.
- Applied object-oriented programming principles throughout the application.
- Implemented form validation to improve data reliability.
- Designed the application for structured record management within a mobile environment.

Repository: <https://github.com/greglie498/studentmanagement>

Smart Campus Navigation System

Lead Engineer / Academic Team Project | 2026

Technology: React, HTML, interactive mapping, UI prototyping

- Served as Lead Engineer on a four-member university project team.
- Coordinated technical development and delegated implementation tasks among team members.
- Contributed to technical decision-making and coordinated the integration of team members' work.
- Developed React-based interface components for the Smart Campus Navigation System.
- Contributed to the interactive campus-map experience.
- Worked on the user journey for searching campus locations and accessing navigation information.
- Assisted with troubleshooting and integration of application functionality.
- Contributed to a mobile-first system concept designed to help students, staff, and visitors locate campus facilities.
- Worked with the team on requirements, use cases, interface design, accessibility considerations, and system prototyping.

Repository: https://github.com/greglie498/Smart_Campus

LEADERSHIP & TEAM EXPERIENCE

Lead Engineer — Smart Campus Navigation System

United States International University–Africa | 2026

- Led the technical development activities of a four-member project team.
- Divided technical tasks according to team members' responsibilities and capabilities.
- Coordinated implementation progress and integration of individual contributions.
- Reviewed and integrated code developed by team members.

- Made technical decisions concerning application functionality and implementation.
 - Contributed directly to React development and the interactive campus-map functionality.
 - Troubleshoot technical issues during development.
 - Supported communication between the technical implementation and the broader project requirements.
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SOFTWARE ENGINEERING EXPERIENCE

Backend Engineering

- Designing RESTful APIs
- Implementing business logic
- Structuring backend applications
- Database integration
- Authentication and authorization
- Third-party API integration
- Transaction workflows
- Validation and error handling
- Testing backend services

Full-Stack Development

- Connecting React/React Native interfaces to backend APIs
- Designing frontend components
- Implementing CRUD workflows
- Managing application state and data flows
- Integrating authentication
- Building database-backed applications

Database Development

- Relational schema design
- PostgreSQL
- MySQL
- MongoDB
- SQLite
- Prisma ORM
- SQL queries
- CRUD operations
- Data validation
- Persistence-layer architecture

Software Architecture

- Clean Architecture
 - Controller-Service-Repository pattern
 - DAO architecture
 - Service Layer architecture
 - Separation of concerns
 - Modular application design
 - Object-oriented design
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REQUIREMENTS & SYSTEM ANALYSIS

Experience gained through academic and software-development projects includes:

- Functional requirements analysis
 - Non-functional requirements definition
 - User-story development
 - Workflow modelling
 - Use-case analysis
 - System architecture planning
 - Database modelling
 - Technical documentation
 - API specification
 - User-interface requirements
 - Accessibility considerations
 - Software project planning
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CERTIFICATIONS & TRAINING

JavaScript Essentials 1

Cisco Networking Academy & OpenEDG JavaScript Institute

LANGUAGES

- **French:** Fluent
- **English:** Fluent
- **Swahili:** Conversational
- **Spanish:** Basic

PROFESSIONAL STRENGTHS

Problem Solving

Strong analytical and debugging approach, with practical experience identifying implementation problems across backend services, databases, APIs, authentication systems, and user interfaces.

Backend Engineering

Particular interest in backend development, application logic, API design, databases, software architecture, authentication, and system reliability.

Technical Leadership

Experience coordinating technical work within a four-member academic development team, delegating tasks, reviewing contributions, integrating code, and making implementation decisions.

Collaboration

Experience working across frontend, backend, database, and documentation responsibilities within academic software-development teams.

Systems Thinking

Ability to consider not only application functionality but also architecture, data flow, security, maintainability, integration requirements, and the environment in which a system will operate.

GITHUB

GitHub: <https://github.com/greglie498>

Selected repositories:

- PanWallet
 - Lost & Found
 - Employee Management System
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CAREER INTERESTS

- Backend Software Engineering
 - Full-Stack Software Engineering
 - API Development
 - FinTech Engineering
 - Mobile Application Development
 - Database Engineering
 - Software Architecture
 - Business Analysis / Technology
 - IT Systems and Technical Support
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PROFESSIONAL DEVELOPMENT AREAS

Current areas of continued development:

- Advanced backend engineering
- Scalable system design
- API architecture
- Cloud technologies
- Cybersecurity
- Database optimization
- DevOps and deployment
- Mobile application development
- Software testing
- FinTech systems
- Business and systems analysis